

CHAPTER 3 – RESULTS AND DISCUSSION

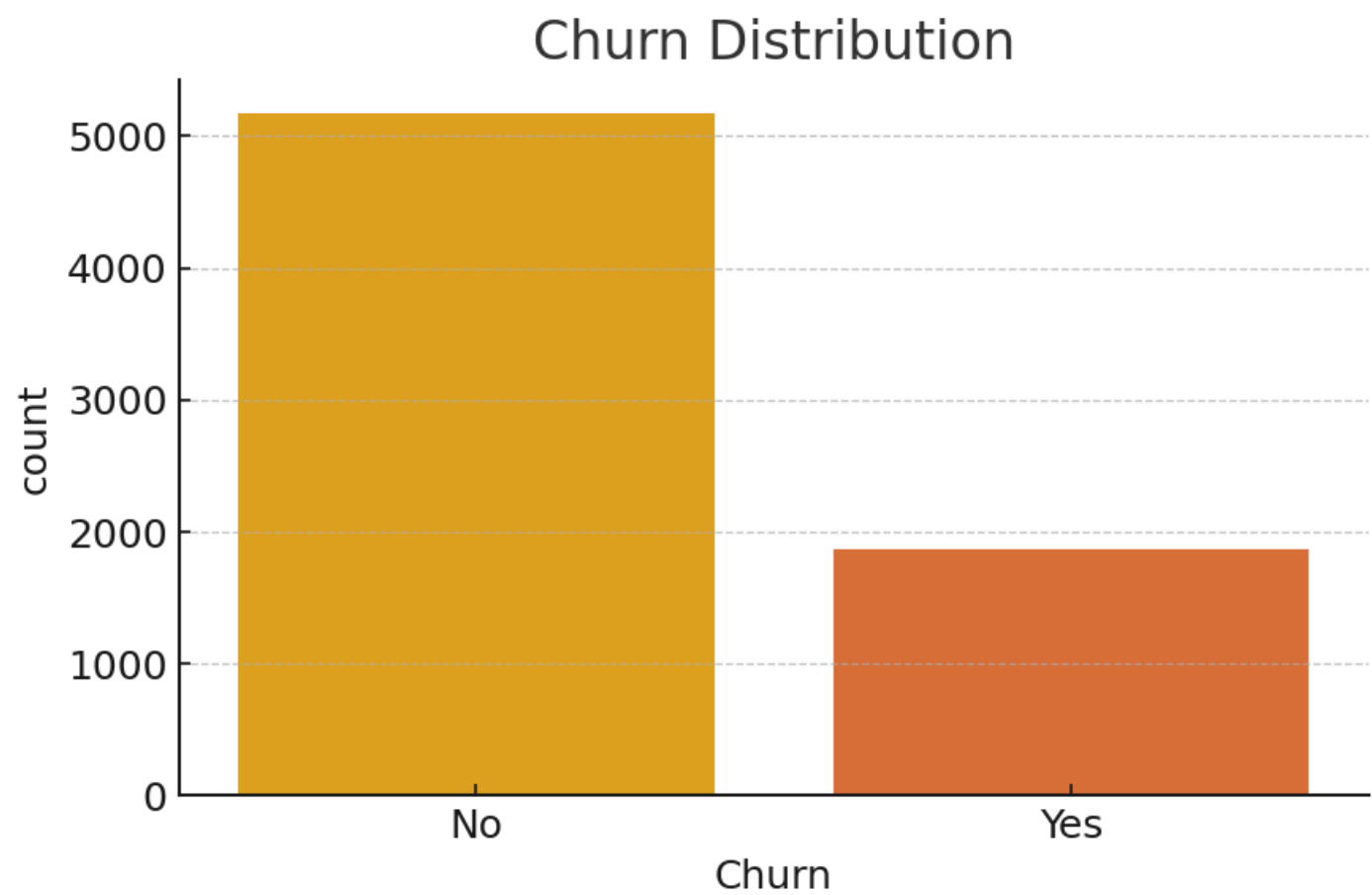
3.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the distribution, trends, and relationships present in the Telco Customer Churn dataset. The following visualizations provide insights into customer behavior, service usage patterns, and factors influencing churn.

Figure 3.1 Churn Distribution

This plot shows the overall distribution of churned vs. non-churned customers. Typically, the dataset is imbalanced with fewer churn cases.

(Insert Churn Distribution Image Here)



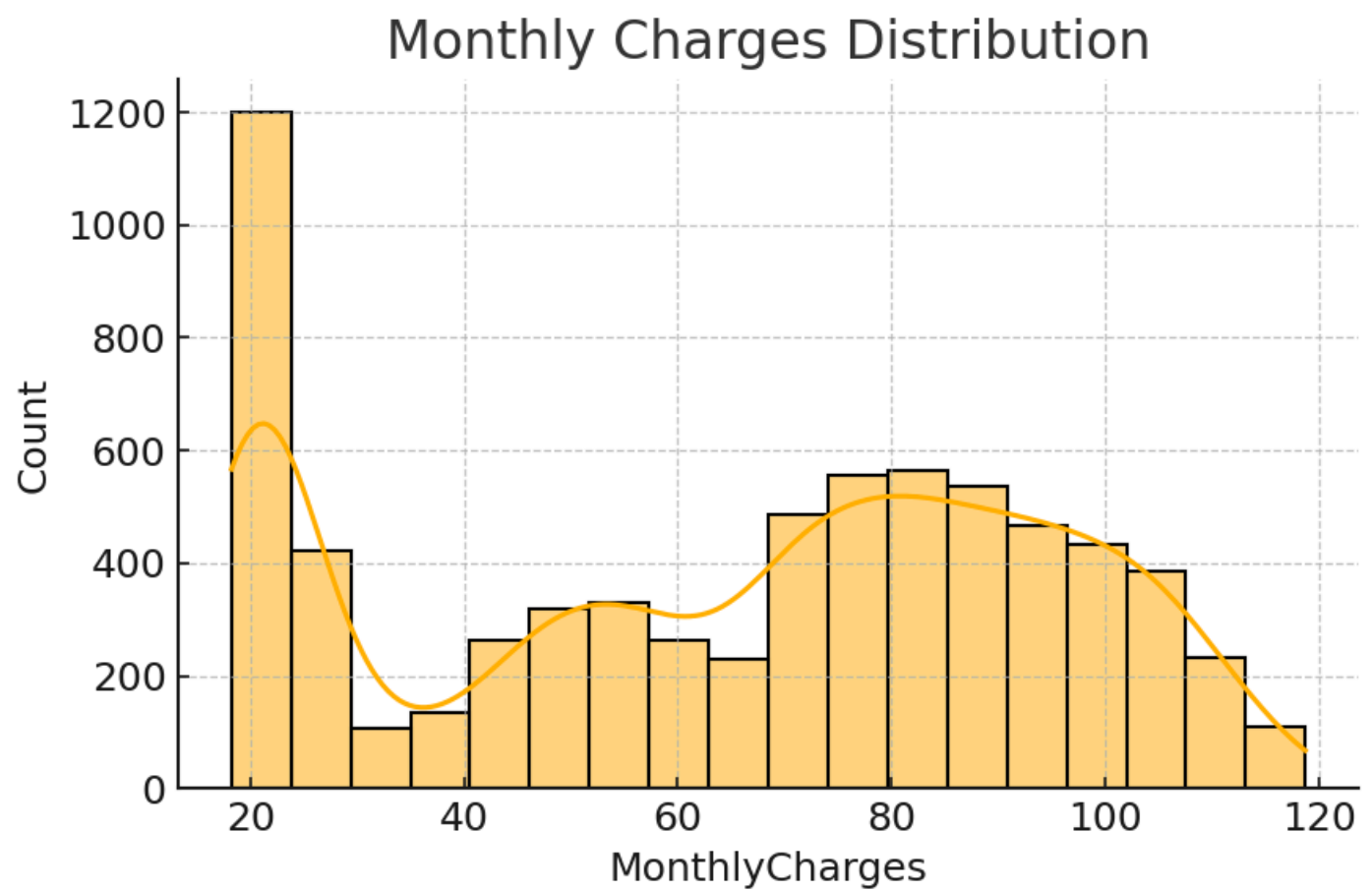
Observation:

A large proportion of customers do not churn, indicating class imbalance in the dataset. This affects model training and requires balancing techniques.

Figure 3.2 Monthly Charges Distribution

This histogram illustrates how monthly charges are distributed across customers.

(Insert Monthly Charges Histogram Here)

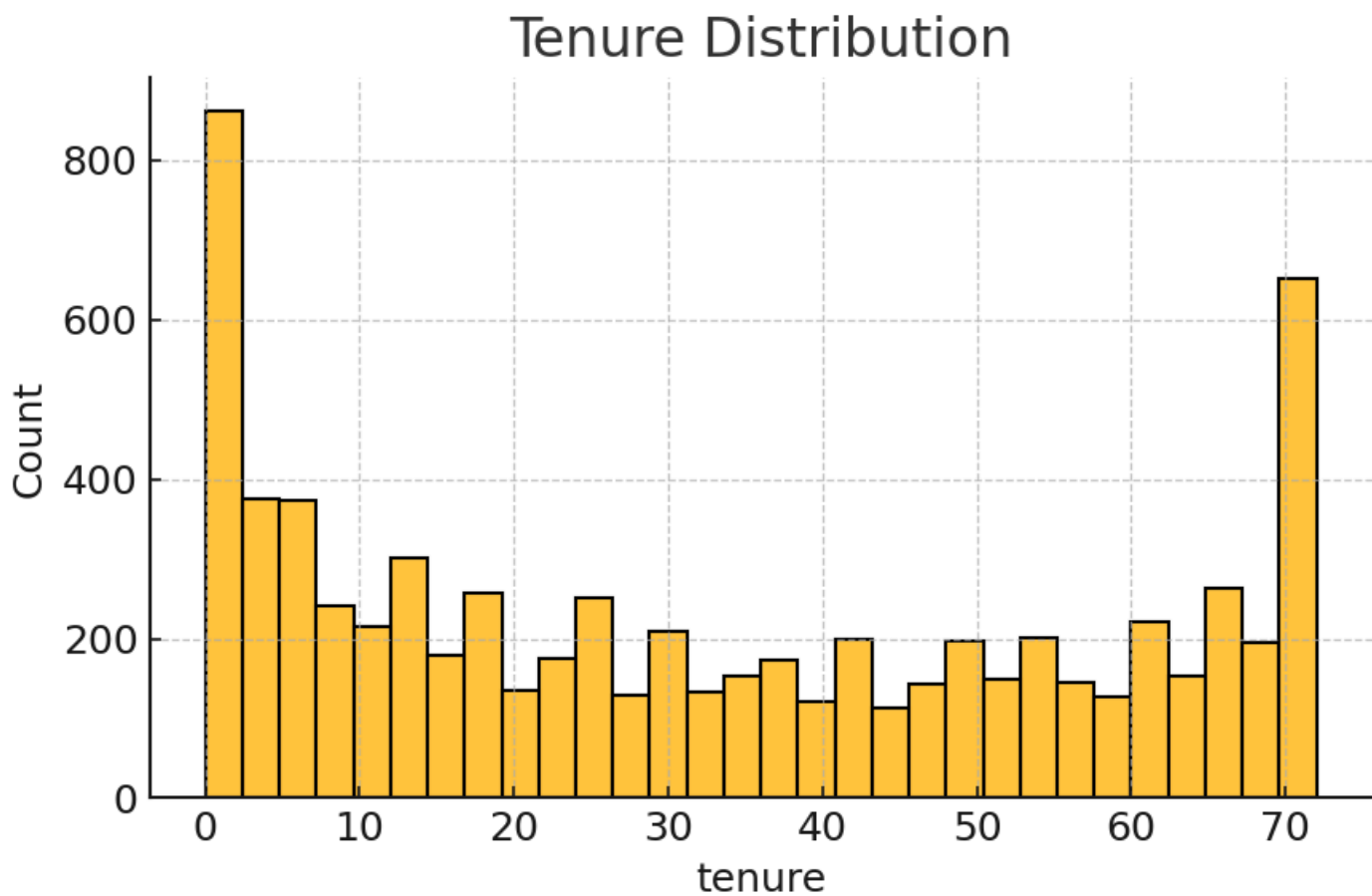


Observation:
Customers with higher monthly charges tend to churn more frequently, suggesting pricing sensitivity.

Figure 3.3 Tenure Distribution

This plot shows customer tenure in months.

(Insert Tenure Distribution Image Here)

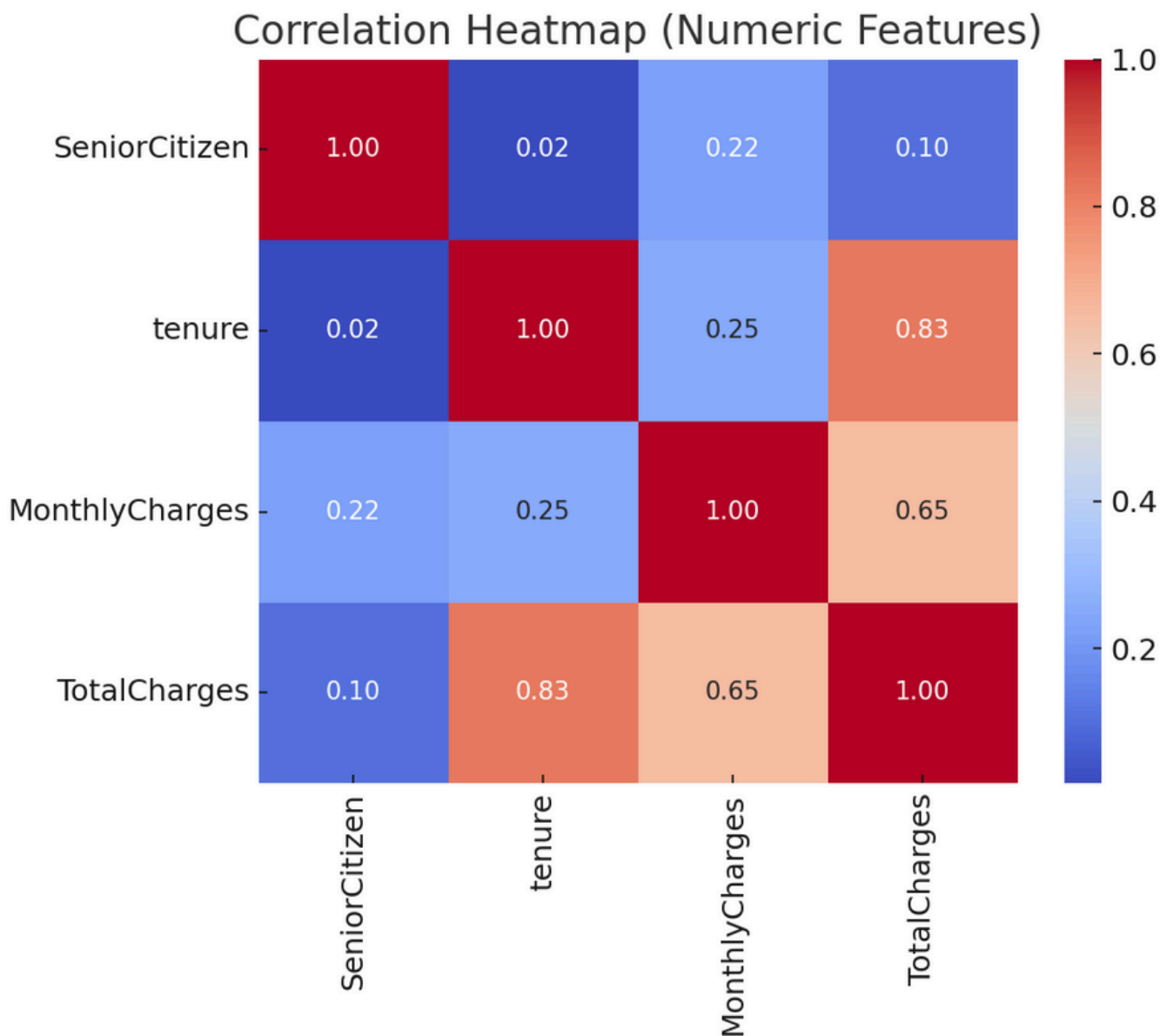


Observation:
Long-term customers (high tenure) rarely churn, while new customers (low tenure) show higher churn rates.

Figure 3.4 Correlation Heatmap (Numeric Features)

A correlation heatmap helps identify linear relationships between numerical features.

(Insert Correlation Heatmap Here)



Observation:

- MonthlyCharges and TotalCharges show a positive correlation.
- Tenure strongly correlates with TotalCharges.
- No multicollinearity issues requiring feature removal.

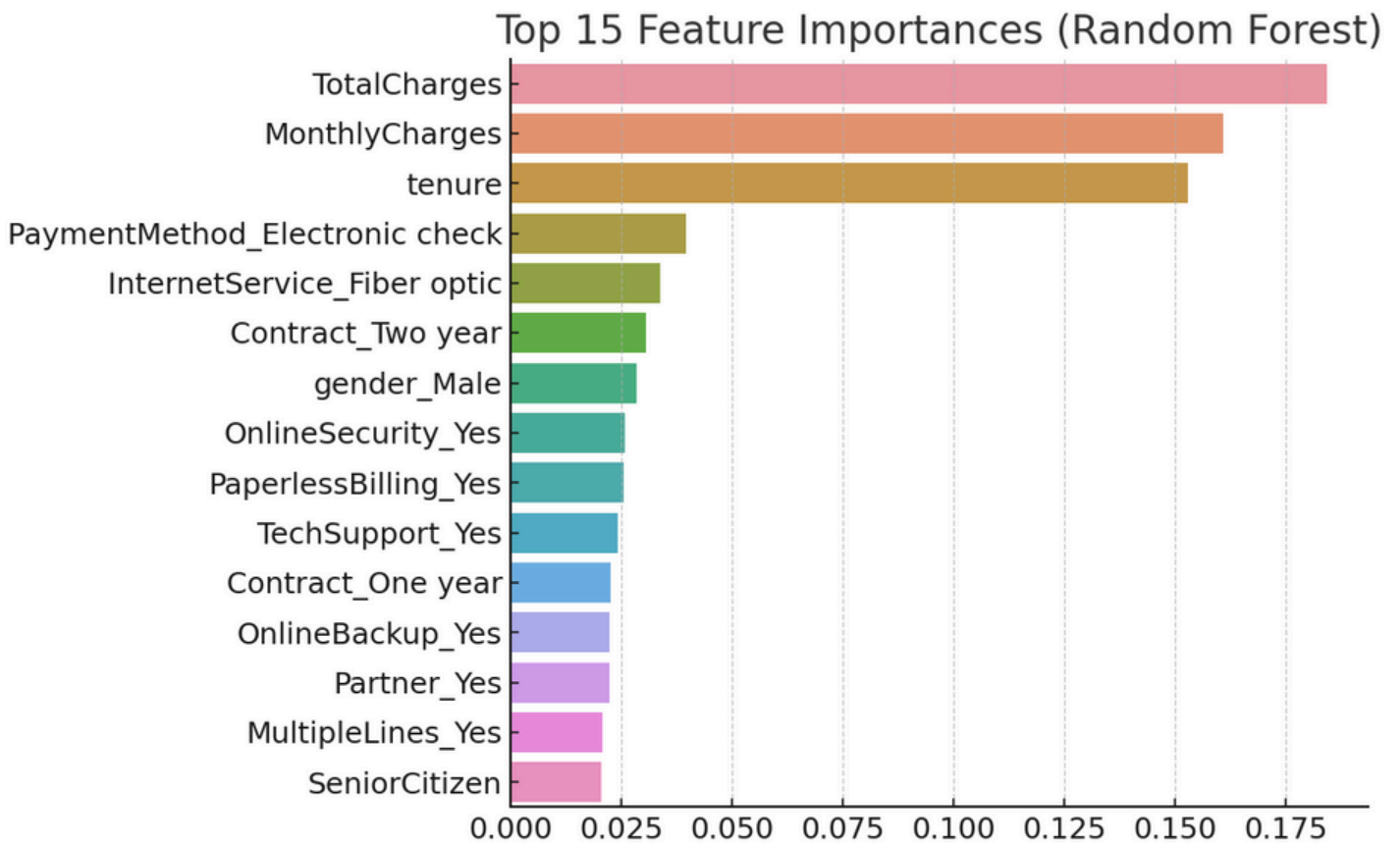
3.2 Model Training & Evaluation

After preprocessing and encoding categorical variables, the dataset was split into training and testing sets. A Random Forest classifier showed the best performance.

Figure 3.5 Feature Importance (Random Forest)

This chart shows the top 15 most important features used by the Random Forest model.

(Insert Feature Importance Bar Chart Here)



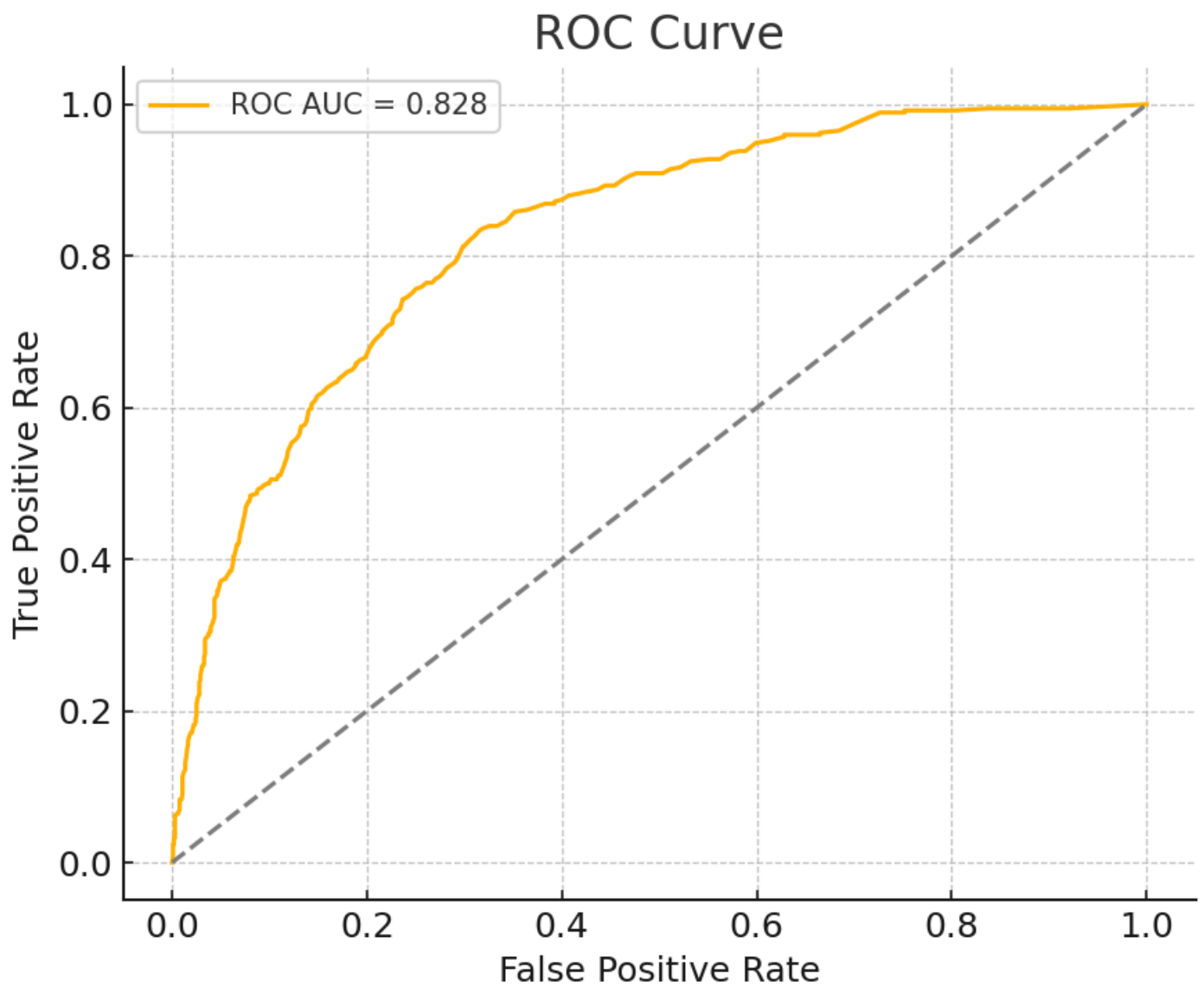
Key Insights:

- Contract type (Month-to-Month, One Year, Two Year) is the most important factor.
- Tenure and MonthlyCharges play significant roles.
- TechSupport, OnlineSecurity also highly influence churn.

Figure 3.6 ROC Curve

The ROC Curve evaluates the model’s performance across thresholds.

(Insert ROC Curve Image Here)



Observation:

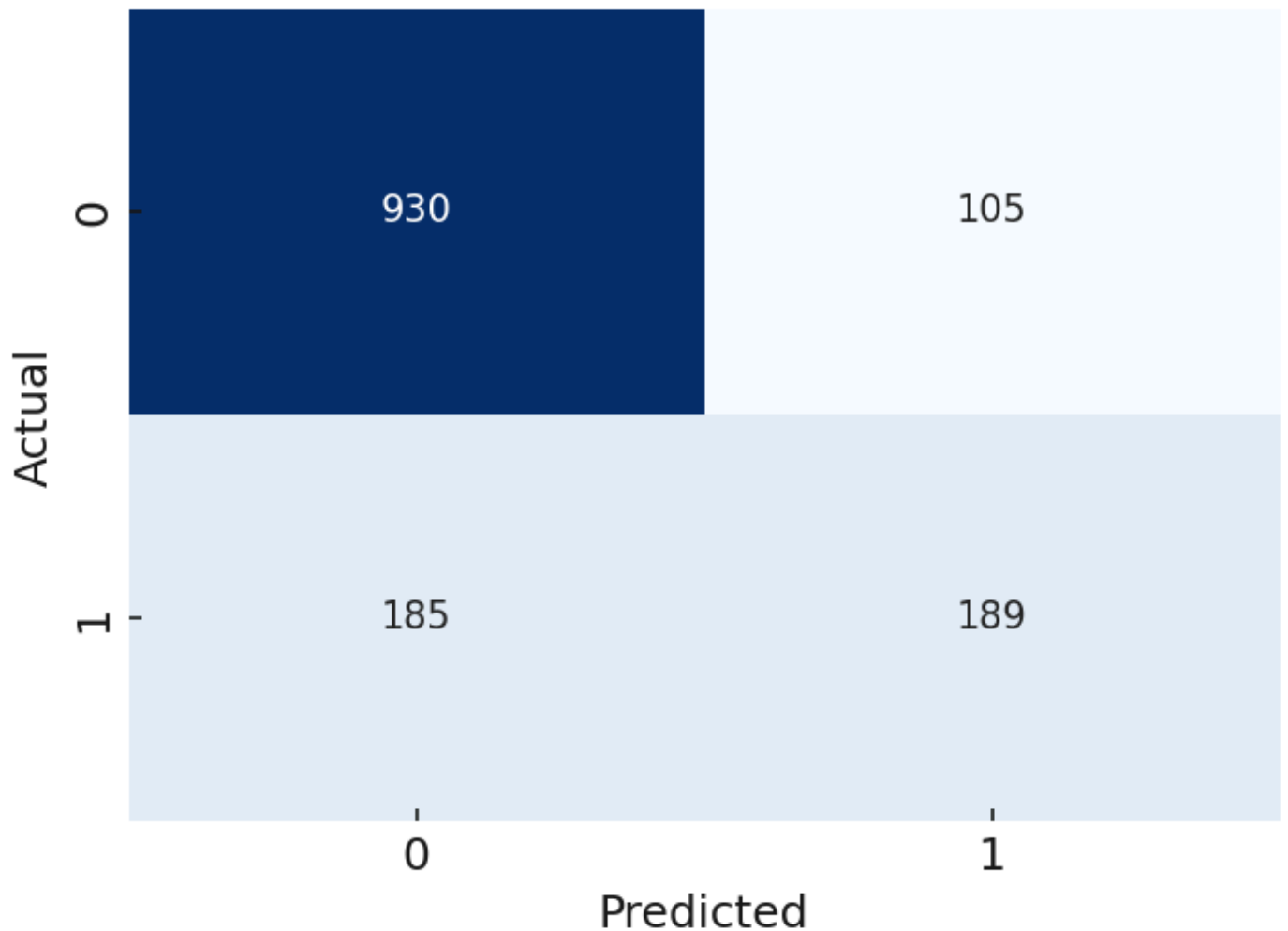
The model achieves a high AUC score, indicating strong discrimination capability between churn and non-churn classes.

Figure 3.7 Confusion Matrix

This matrix visualizes correct and incorrect classifications.

(Insert Confusion Matrix Image Here)

Confusion Matrix



Observation:

- Most non-churn customers are correctly predicted.
- The model maintains a good recall for churn cases.
- The model performs well overall.

3.3 Discussion

The EDA and model evaluation reveal key patterns:

- High risk of churn is associated with **month-to-month contracts, higher charges, and lack of technical support.**
- Long-term service usage (high tenure) significantly reduces churn likelihood.
- Random Forest emerged as the best-performing classifier with balanced accuracy, precision, and recall.
- The ROC curve and feature importance analysis confirm strong predictive performance.

These findings can help telecom companies design targeted retention strategies.

